Patterns in whitebark pine regeneration and their relationships to biophysical site characteristics in southwest Montana, central Idaho, and Oregon, USA

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Abstract: Declines of whitebark pine (*Pinus albicaulis* Engelm.) have occurred across much of the species' range over the last 40 years due to mountain pine beetle outbreaks and white pine blister rust infection. Management efforts to stem these declines are increasing, yet the long-term success of whitebark pine depends on the species itself adapting to the modern environment. Natural regeneration will be a critical part of this process. We examined patterns in natural whitebark pine regeneration as related to the biophysical environment in sixty 0.1 ha plots in Montana, Idaho, and Oregon. Whitebark pine regeneration was present in 97% of our plots and varied widely in density from 0 to 17 000 seedlings/ha and 0 to 2680 saplings/ha. Using nonparametric correlation analysis and ordination techniques, we found whitebark pine regeneration abundance was unrelated to stand age but significantly related to several biophysical site characteristics, including positive relationships with elevation and canopy tree mortality caused by mountain pine beetle and negative relationships with moisture availability, temperature, and subalpine fir importance. Our findings indicate that whitebark pine is regenerating in many areas and that the widespread mortality from recent mountain pine beetle outbreaks may provide suitable settings for whitebark pine regeneration given sufficient seed sources.

Résumé : Des déclins du pin à écorce blanche (*Pinus albicaulis* Engelm.) sont survenus dans presque toute son aire de répartition au cours des 40 dernières années à cause des infestations du dendroctone du pin ponderosa et des infections de la rouille vésiculeuse du pin blanc. Les efforts d'aménagement pour enrayer ces déclins augmentent, mais à long terme, le succès du pin à écorce blanche dépend de son adaptation à l'environnement moderne. La régénération naturelle sera un élément crucial de ce processus. Nous avons étudié les patrons de la régénération naturelle du pin à écorce blanche en fonction de l'environnement biophysique dans 60 placettes de 0,1 ha dans les états du Montana, de l'Idaho et de l'Oregon, aux États-Unis. La régénération du pin à écorce blanche était présente dans 97 % des placettes et sa densité variait énormément, soit de 0 à 17 000 semis/ha et de 0 à 2680 gaules/ha. En utilisant des techniques d'analyse de corrélation non paramétrique et d'ordination, nous avons établi que l'abondance de la régénération de pin à écorce blanche n'était pas reliée à l'âge du peuplement, mais qu'elle était reliée à plusieurs caractéristiques biophysiques de la station, incluant des relations positives avec l'altitude et avec la mortalité des arbres dominants causée par le dendroctone du pin ponderosa et des relations négatives avec la disponibilité en eau, la température et l'abondance du sapin subalpin. Nos résultats indiquent que le pin à écorce blanche se régénère à plusieurs endroits et que la forte mortalité causée par les récentes infestations du dendroctone du pin ponderosa peut créer des conditions adéquates pour l'établissement de sa régénération si les sources de semences sont suffisantes.

[Traduit par la Rédaction]

Introduction

The continued persistence of whitebark pine (*Pinus albicaulis* Engelm.) as an important species in high-elevation forests of western North America is a topic of major concern among land managers and researchers (Tomback et al. 2001*b*). Dramatic declines have been observed in the health and dominance of whitebark pine across the species' range over the last 40 years (Kendall and Keane 2001). These declines have been attributed to the invasive white pine blister rust (*Cronartium ribicola* (A. Dietr.) J.C. Fisch.), advancing succession as a result of fire suppression, and episodic mountain pine beetle (*Dendroctonus ponderosae* Hopkins) outbreaks over the 20th century. Concern over the deterioration of whitebark pine forests is based on the potential cascade of ecological effects that may result because of the critical role the species fills in subalpine forest communities through enhanced biodiversity; through regulation of hydrology and watershed dynamics; and as the foundation of an ecosystem involving Clark's nutcrackers (*Nucifraga columbiana* Wilson), red squirrels (*Tamiasciurus hudsonicus*

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Trouessart), grizzly bears (*Ursus arctos* L.), and black bears (*Ursus americanus* Pallas) (Ellison et al. 2005).

In response to these declines, active management and restoration projects in whitebark pine forest are becoming increasingly common, yet to persist over the long term the trees themselves must adapt to changes in the biotic and abiotic environment, particularly with respect to blister rust. Change and adaptation in species and natural systems occurs through selective processes that operate at the scale of generations, and therefore successful whitebark pine regeneration will be one critical component of the long-term success of this species.

Early research on whitebark pine regeneration focused primarily on the mutualistic relationship between whitebark pine and its primary dispersal agent, the Clark's nutcracker, and on the ecological and evolutionary implications of this relationship, particularly with respect to the behavioral tendencies of Clark's nutcrackers when they create seed caches (Lanner 1982; Tomback 1982). Clark's nutcrackers preferentially cache whitebark pine seeds in open and recently disturbed settings, and most studies of the patterns and abundances of whitebark pine regeneration have focused on postfire settings (e.g., Tomback et al. 1993, 1995, 2001*a*). Clark's nutcrackers have also been documented caching whitebark pine seeds in a variety of other microsites (Hutchins and Lanner 1982), and recent research identified variable levels of advanced whitebark pine regeneration across a wide spectrum of biophysical settings, including stands that had experienced high levels of mortality related to 20th century mountain pine beetle outbreaks (Mellmann-Brown 2005; Moody 2006).

The community dynamics and spatial patterns of animal-dispersed plants are highly influenced by the nonrandom behaviors of their dispersalists (Jordano et al. 2007), yet few assessments of patterns in whitebark pine regeneration at the landscape-scale have been conducted (but see Zeglen 2002). A more complete understanding of where whitebark pine regeneration is likely to occur and succeed would enhance management and restoration activities in whitebark pine communities, particularly with respect to site selection for the outplanting of blister-rust-resistant whitebark pine seedlings (Hoff et al. 2001) and the application of prescribed fire (Keane and Arno 2001). In response to this need, our study seeks a better understanding of the spatial patterns in natural whitebark pine regeneration and how these patterns relate to the biophysical environment. Our guiding questions for this research were (1) Is whitebark pine regenerating in our study area?, (2) How do patterns of whitebark pine regeneration vary across the landscape and does this variability relate to the biophysical environment?, and (3) How do patterns of whitebark pine regeneration relate to past disturbances and in particular mountain pine beetle outbreaks?

Study area

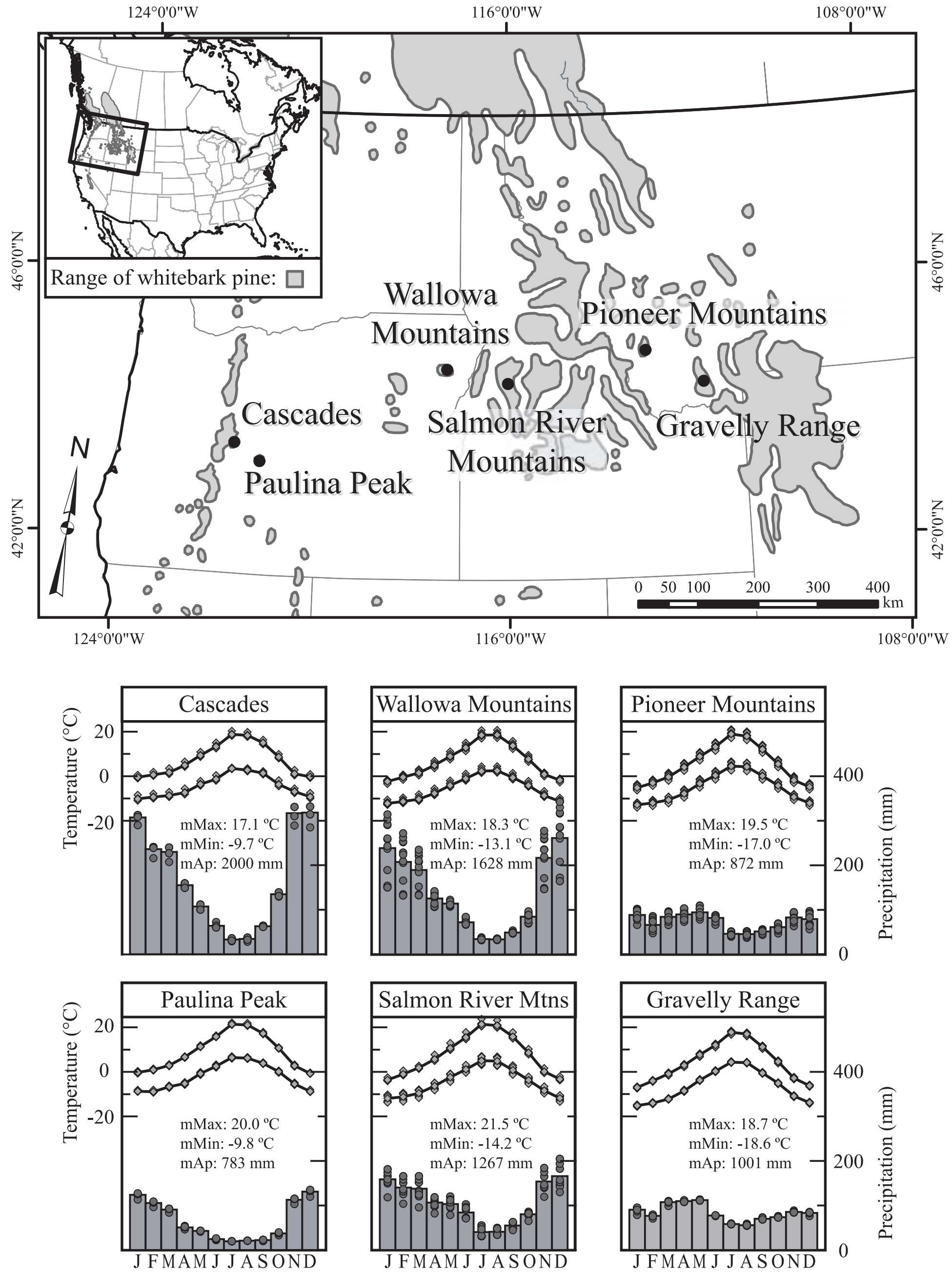
Whitebark pine is widely distributed across the western United States, with the native range of the species spanning 36°–55°N and 108°–127°W. Within this range, whitebark pine habitat is found from the lower subalpine zone to the upper forest limits and timberline (Arno and Hoff 1990). Our study area extends from southwest Montana to western Oregon between ca. 42° and 45°N, with study sites in the Gravelly Range and Pioneer Mountains of Montana, the Salmon River Mountains of Idaho, and the Wallowa Mountains, Paulina Peak, and Cascade Range of Oregon (Fig. 1). These mountain ranges are located within the Beaverhead–Deerlodge National Forest, the Payette National Forest, the Wallowa–Whitman National Forest, and the Deschutes National Forest, respectively.

Topographic diversity in mountain environments strongly influences ecological patterns and processes. The diverse character of the different mountain ranges included in our study area provided us with a picture of whitebark pine regeneration across a broad array of physical settings. The Gravelly Range contains some steep and rugged slopes, but our study sites were located on the broad crest of the range that exists predominantly above 2900 m and has generally rolling topography with gentle slopes. In contrast, the Pioneer Mountains, Salmon River Mountains, and Wallowa Mountains are each an extensive mountain massif and highly dissected by glacial valleys and ridges, and our study sites were situated in areas of steep slopes and sharp environmental gradients. Our sites in the Cascades were located on the south and southwest flanks of Mount Bachelor, a relatively young volcanic mountain with a regular conical shape, and at the crest of Black Crater, an irregularly shaped cinder cone. Paulina Peak is the remnant of Newberry Volcano, and our sites were situated along a high-elevation rim bordering the south and east sides of what is now Newberry Caldera.

Broad similarities and a few distinct differences existed in the annual patterns of temperature and precipitation at our sites based on site-specific climate data obtained from the PRISM data set (PRISM Group, Oregon State University, http://www.prismclimate.org, created 13 Nov 2008) (Daly et al. 2008). In general, the sites shared a common pattern of drier summers and wetter winters, although the peak in winter precipitation was much more pronounced in the Cascades, Wallowa Mountains, and Salmon River Mountains than at the other sites (Fig. 1). Mean total annual precipitation of the different mountain ranges was affected by regional-scale rain shadows and ranged from 783 mm across the sites on Paulina Peak up to 2000 mm at the Cascades sites, whereas mean monthly maximum and minimum temperatures exhibited a gradient in extremes from the western, maritime sites to the eastern, continental sites (Fig. 1). Variability in the climate conditions within each landscape was highly influenced by the general topography of the mountain ranges, with more variable topography and climate in the Pioneer Mountains, Salmon River Mountains, and Wallowa Mountains contrasting with less variability in the Gravelly Range, Paulina Peak, and the Cascades (Fig. 1).

Our study sites were located in the upper forest zones of each mountain range where whitebark pine was the dominant or codominant tree species (Table 1). In addition to whitebark pine, several other tree species occurred in at least one site, including subalpine fir (*Abies lasiocarpa* (Hook.) Nutt.), Engelmann spruce (*Picea engelmannii* Parry ex Engelm.), lodgepole pine (*Pinus contorta* Douglas ex Louden), Douglas-fir (*Pseudotsuga menziesii* (Mirb.) Franco), and mountain hemlock (*Tsuga mertensiana* (Bong.) Carr.). Herbaceous communities varied considerably among the mountain ranges and among the individual sites within each range because of

Fig. 1. Study area showing the range of whitebark pine (gray shading), location of studied mountain ranges (black dots), and climographs for each mountain range. The climographs represent the mean monthly maximum, mean monthly minimum, and mean monthly precipitation for each plot as scatter points with the overall means of each of these variables for each mountain range indicated by the line graphs (temperature) and the bar graphs (precipitation). The data were derived from PRISM climate grids for the United States (PRISM Group, Oregon State University, http://www.prismclimate.org, created 13 Nov 2008) (Daly et al. 2008).



differences in the slope, aspect, substrate, and topographic position of each site. The dominant species based on percentage of ground cover included *Arnica alpine* (L.) Olin, *Juniperus* spp., *Lupinus* spp., *Penstemon* spp., *Phlox hoodii*, *Ribes* spp., *Phyllodoce empetriformis* (Sm.) D. Don, *Vaccinium scoparium* Leiberg ex Coville, and *Salix* spp.

Methods

Data collection and processing

We used a geographic information system to identify the geographic centroid of 60 upper-elevation stands among the different mountain ranges that contained whitebark pine based on geospatial USDA Forest Service stand inventory data provided by each National Forest. We viewed aerial photographs to confirm the vegetation type at each plot. These stands existed on a variety of slopes and aspects and were near the tree line, yet still appeared to have relatively continuous canopies. The Universal Transverse Mercator coordinates of each stand centroid were marked on field maps and programmed into a handheld global positioning system device.

Table 1. Site setting, stand structure, and stand compositional information for each mountain range.

					Importance values				
Range	*n*	Elevation (m)	Density (trees/ha)	Basal area (m^2/ha)	*Pinus albicaulis*	*Abies lasiocarpa*	*Picea engelmannii*	*Pinus contorta*	*Tsuga mertensiana*
Cascades	6	2330±41	802±130	31±4	179±6	6±4	0±0	0±0	14±7
Gravelly Range	8	2887±27	848±120	34±4	176±12	22±11	2±1	0±0	0±0
Paulina Peak	7	2306±29	976±165	36±5	112±20	0±0	0±0	74±21	13±6
Pioneer Mountains	14	2728±27	1051±177	38±5	152±12	2±1	9±5	31±12	0±0
Salmon River Mountains	12	2369±26	378±38	30±4	112±13	74±11	2±1	11±7	0±0
Wallowa Mountains	13	2348±35	519±45	25±3	92±14	73±10	4±2	31±15	0±0

Note: Values are means ± 1 SD.

In the field, we found that two of the stand centroids were located on cliff edges or other unsafe slopes, four were in forest openings, one was approximately 25 m offshore in a lake, and one was on the margins of a treeless wetland. In these cases, we adjusted the plot location 100 m in a randomly chosen direction. In addition, two of the larger stands in the Wallowa Mountains appeared relatively heterogeneous in disturbance history, structure, and composition. We therefore used aerial photographs, stand inventory maps, and the global positioning system device to randomly select up to two additional sites within these stand boundaries.

A 0.1 ha circular plot was placed at the centroid or adjusted centroid of each stand to collect site characteristic and forest demographic information. We recorded the elevation, average slope, aspect, topographic position, microsite characteristics, and substrate type for each plot. The species, diameter at breast height, canopy class, and health of all trees were recorded within the 0.1 ha plot. Canopy class was defined as canopy (≥50% of the tree canopy exposed to direct overhead light) and subcanopy (<50% of the tree canopy exposed to overhead light). Tree health was subjectively categorized as alive (healthy canopy, fully capable of reproduction), declining (partial canopy dieback but still capable of producing cones, or the presence of pitch tubes caused by active mountain pine beetle infestation), and dead. All living whitebark pine trees were visually searched for evidence of blister rust infection, including active cankers, inactive cankers, rodent chew, and flagging branches. We did not survey dead whitebark pine trees for blister rust because of the tendency of most dead trees in our plots to have lost their small branches and sloughed off their bark, greatly reducing our ability to identify past infections. All dead trees were inspected for physical damage, char, and j-shaped mountain pine beetle galleries to determine the cause of death.

We tallied all saplings (≥2 cm diameter at ground level and <5 cm diameter at breast height) by species and all seedlings (<2 cm diameter at ground level) by species within nested plots of 0.05 and 0.01 ha, respectively. Increment cores were collected along two radii as low on the bole as possible from all living and dead trees ≥5 cm diameter at breast height within the 0.05 ha plot to determine stand age and disturbance history and to detect the presence of blue stain fungus, an indicator of mortality caused by mountain pine beetle. Additional cores were taken through the scar face and healing lobes of any trees in the coring plot that displayed basal or strip-kill scars to date the event that caused the scar.

All of the increment cores were air-dried, glued into core mounts, and sanded using progressively finer grit sand paper until individual xylem cells were clearly visible under 7–45× magnification. We crossdated each core with master chronologies developed for each mountain range using visual and graphical methods (Stokes and Smiley 1996). The inner dates of cores that did not reach the pith of a tree but exhibited sufficient curvature in the inner rings were corrected using pith estimators based on concentric circles and constant growth rates. Cores that were rotten near the center and did not contain pith or sufficient curvature to estimate the rings to pith were assigned minimum ages and excluded from our age-structure analyses. Cores taken from scarred trees were examined in the context of multiple lines of evidence, including field notes on the presence or absence of charcoal, stand structure data, timing of other injuries in the same plot, and presence or absence of blue stain at the scar face, to assign a cause to the injury (physical abrasion, mountain pine beetle strip kill, fire, or unknown disturbance).

We summarized the age-structure data into 10 year age classes and calculated a number of disturbance-related metrics. Minimum stand age was determined as the age of the oldest tree in the plot and was used as a measure of the time since the last stand-replacing disturbance. We identified cohorts where the sum of trees that established over any 30 year period (three consecutive age-structure bins) was ≥5 trees and ≥25% of the total number of cored trees in the plot and used the time since the last cohort establishment as an estimate of the time since the last disturbance, stand replacing or not. We assigned cohort establishment dates, fire scar dates, and mortality events related to mountain pine beetle to decadal bins and used these to calculate a decadal disturbance frequency for each plot.

We collected site-specific climate data for each of our plots. The climate data we used were sampled from the Spline climate data set developed by Rehfeldt (2006) and from the PRISM climate data developed by the PRISM Group at Oregon State University (Daly et al. 2008). The climate variables we obtained included mean monthly maximum and minimum temperatures and mean monthly precipitation from the PRISM data, and mean annual precipitation and mean annual temperature from the Spline data set. We seasonalized the monthly variables into spring (MAM), summer (JJA), fall (SON), and winter (DJF) variables and included a number of derived variables related to the length of the growing season and the number of degree-days (Table 2). Following Rehfeldt et al. (2008), we also calculated a growing season dryness index as the ratio of total summer precipitation to the degree-days >5 C° that accumulate during the frost-free season based on the Spline data (Table 2). Lower values of the growing season dryness index indicate sites that are more likely to experience drought conditions over the course of a year.

We calculated a suite of standard forest metrics (frequency, relative frequency, basal area, relative basal area, and importance values) for each species present and stratified by canopy class and health category (Table 2). In addition to the raw variables of slope, aspect, and elevation, we converted plot aspect to a linear metric, calculated relative elevation, and determined a topographic relative moisture index (TRMI) and a relative TRMI for each plot. Linear aspect was calculated as 1 + cos[π(aspect – 45)/180] and resulted in a value from 0 (warmer, drier southwest-facing slopes) to 2 (cooler, moister northeast-facing slopes). Relative elevation was calculated by subtracting the elevation of the lowest plot within a mountain range from the elevation of each of the other plots and represented an estimation of the elevational position of each stand within the distribution of sampled whitebark pine stands in each of the mountain ranges. The TRMI was developed to quantify the effects of stand-scale topography on effective moisture availability in mountainous landscapes (Parker 1982). TRMI values range from 0 to 60 and were calculated as the sum of values assigned to slope steepness (0–10), slope configuration (0–10), slope aspect (0–20), and slope position (0–20). Lower numbers indicate sites with less moisture availability and higher numbers indicate sites with greater moisture availability. To enhance our abilities to use the TRMI in comparisons among mountain ranges, we calculated a rTRMI for each site as follows:

$$[1] \qquad \mathrm{rTRMI}_s = \frac{\mathrm{TRMI}_s \times \mathrm{MAP}_s}{\frac{1}{n}\sum_{i=n}^{n}\mathrm{TRMI}_i}$$

where $\mathrm{TRMI_s}$ and $\mathrm{MAP_s}$ are the TRMI and mean annual precipitation (PRISM data) for site *s*, respectively, and TRMI_i is the TRMI for the *i*th site. This measure incorporates site-specific precipitation and topography to produce a measure of effective moisture availability that is comparable across sites with similar topography but different climate regimes.

Table 2. Biophysical variables calculated for each site including derived climate variables and stand and site metrics calculated for each site.

Code	Description
Derived climate variable	
D100	Day of year the sum of degree-days reaches 100
DD0	Degree-days <0 °C
DD5	Degree-days >0 °C
FFP	Day of year of the first freezing date of autumn
FDAY	Length of the frost-free period
GSDD5	Degree-days >5 °C accumulating in the frost-free period
GSP	Growing season precipitation (April to September)
MMAX	Mean maximum temperature in the warmest month
MMIN	Mean minimum temperature in the coldest month
MTCM	Mean temperature in the coldest month
MTWM	Mean temperature in the warmest month
SDAY	Day of year of the last freezing date of spring
ADI	Annual dryness index (DD5/MAP)
SMI	Summer dryness index (GSDD5/MAP)
GDI	Growing season dryness index (GSP/DD5)
Calculated stand metric	
Freq	Frequency[a]
RF	Relative frequency[a]
BA	Basal area (m^2)[a]
rBA	Relative basal area[a]
IV	Importance values ([RF + rBA]×100)[a]
lASP	Linear aspect (Cosine[Aspect+45])
Elev	Elevation
rElev	Relative elevation (see text for definition)
TRMI	Topographic relative moisture index
rTRMI	Relative topographic relative moisture index (see text and eq. 1 for definition)
Min Age	Minimum age of stand
DDF	Decadal disturbance frequency
TSC	Time since cohort establishment

Note: MAP, mean annual precipitation.

[a]Calculated for each permutation of species present, canopy class (C = canopy, S = subcanopy), and health category (A = alive, DEC = declining, D = dead).

Data analysis

We used two nonparametric tests and ordination analyses to explore the potential relationships between whitebark pine regeneration and site biophysical characteristics across our study area. We used Kruskal–Wallis' one-way analysis of variance by ranks to assess the overall similarity in levels of whitebark pine regeneration among the different mountain ranges and Kendall's τ correlation analyses to describe the strength of the relationships among individual biophysical site characteristics and whitebark pine regeneration.

We used both unconstrained (Principal Coordinates Analysis; PCO) and constrained (Canonical Correlation Analysis; CCorA) ordinations to address the multicollinearity in our environmental variables and to further assess the relationships identified in our correlation analyses. Because of the similarity in the results of our Kendall's τ correlation analyses comparing seedling and sapling densities to biophysical site characteristics, we used only the variables identified as significant with respect to sapling density to create a matrix **X** that we ordinated using Canonical Analysis of Principal Coordinates (CAP). CAP is conducted by first using PCO on any type of dissimilarity matrix or distance measure, and then conducting CCorA on the axes of the PCO (Anderson and Willis 2003). This approach allows for flexibility in the choice of dissimilarity or distance measures, and by conducting the CCorA on the PCO axes produces less arbitrary results for the CCorA. Using this approach, CAP has been shown to be effective in identifying ecological patterns in multivariate data that are otherwise masked in the results of unconstrained ordinations (Anderson and Willis 2003). Matrix **Y** for the CCorA was composed of whitebark pine seedling and sapling densities. Our ordinations were based on a symmetrical Gower dissimilarity matrix due to the different states of the biophysical variables. All ordinations were conducted using the program CAP v. 12 (Anderson and Willis 2003), with the program determining the number of PCO axes (m) to include in the canonical analysis by sequentially increasing the number of m and each time calculating the residual error. The number of axes resulting in the minimum residual error is chosen, with the test run on 9999 random permutations.

Results

We inventoried 1240 living seedlings, 3051 living saplings, and 4176 trees, of which we cored 2346, in 60 plots during the summers of 2006–2008 (Table 3). Of these, 1004, 1546, and 2666 were whitebark pine, respectively. We observed whitebark pine regeneration in 97% (n = 58) of our plots, with seedling densities ranging from 0 to 17 000/ha and sapling densities ranging from 0 to 2680/ha (Table 3). Increment cores were collected along two radii from 2346 trees, and we were able to assign accurate inner dates or pith estimations to 87% (n = 2046) of the cored trees. The majority of the undated cores were collected from trees with rotten centers. Pith was included in 16% (n = 365) of the cores and the average correction for cores that did not include pith was 7 ± 5 SD years (range: 1–30 years). Stand setting, structure, and composition varied widely among the mountain ranges (Tables 1 and 3). Overall, blister rust infections were observed on 37% (n = 660) of the living whitebark pine trees we inventoried, with rates of infection in individual plots ranging from 0% to 100%. Out of the 2666 whitebark pine trees inventoried, 33% (n = 887) were dead. Mountain pine beetle activity was the primary cause of mortality, with 83% (n = 619) of all dead whitebark pine exhibiting extensive pitch tubes, beetle galleries, and (or) blue stain fungus. We found evidence of past fires in 35% (n = 21) of our plots in the form of fire-scarred trees and (or) postfire cohorts. In general, disturbance frequency was highest in the Salmon River Mountains and lowest in the Wallowa Mountains (Table 3).

Whitebark pine seedling and sapling abundances varied significantly among the six mountain ranges (seedlings: H = 33.29, P < 0.001; saplings: H = 33.66, P < 0.001) (Table 4). Our correlation analyses identified a number of significant relationships between biophysical site characteristics and both seedling and sapling abundance (Table 5). The strongest relationship for both categories of regeneration was an inverse relationship with the importance of subalpine fir. Both seedlings and saplings were positively correlated with the density of whitebark pine as well as the density of dead trees and whitebark pine killed by mountain pine beetle. Whitebark pine regeneration was positively correlated with overall stand density and the density of lodgepole pine and mountain hemlock. Elevation and whitebark pine regeneration was positively correlated, with relative elevation showing a stronger relationship than absolute elevation. Several climate variables were significantly related to both seedling and sapling abundance (Table 5), illustrating the general pattern of greater whitebark pine regeneration at the cooler, drier sites of our study area. The direction of these relationships was similar between seedlings and saplings, but the strength of the relationships was greater with respect to sapling abundance in almost all cases. Whitebark pine sapling density was also weakly but significantly related to the disturbance frequency of our sites.

The CAP results refined the patterns we identified in our correlation analyses. The first two axes of the unconstrained PCO ordination explained 68% of the variance in the biophysical variables significantly related to whitebark pine sapling density. A bi-plot of the axis scores illustrated systematic differences in the multivariate structure of the biophysical variables among the mountain ranges, indicating that each range has a unique physical and climatic envelope (Fig. 2a). The constrained CCorA ordination was based on the first 5 axes of the PCO, which together explained 94% of the variance in the biophysical variables. The CCorA generally agreed with our correlation analyses (Table 5) and highlighted the underlying similarities in the relationships among site biophysical characteristics and whitebark pine regeneration across our study area (Fig. 2b). Whitebark pine seedling abundance was correlated at –0.54 with CCorA axis 1 and –0.26 with CCorA axis 2 while sapling abundance was correlated at –0.59 and 0.19, respectively (Fig. 2b). In the context of the CCorA ordination, these results indicate that both seedlings and saplings are generally more abundant at the sites that were higher, drier, and colder with denser forests, more whitebark pine, and higher levels of mountain pine beetle mortality relative to our data set as a whole, and are less abundant at warmer, wetter sites with greater subalpine fir importance. These patterns hold both within and among the different mountain ranges. The different relationships between seedling and sapling abundance and CCorA axis 2 indicate that although more whitebark pine regeneration occurred in the relatively colder sites across the landscape, within the context of this pattern seedling density was greater on warmer sites while sapling density was greater at the cooler sites.

Discussion

The presence of early and advanced whitebark pine regeneration

The nearly ubiquitous presence of whitebark pine seedlings and saplings across our study area was encouraging, as past landscape-scale assessments of whitebark pine have reported patchy occurrences of regeneration or abundant seedlings with few saplings present (Zeglen 2002). The presence of seedlings at nearly all of our sites reflects two consecutive mast years in 2005 and 2006 that occurred across our study area (Bob Keane, Research Ecologist, USDA Fire Sciences Laboratory, personal communication, 2007). Whitebark pine seed germination often lags by 1–2 years when the seeds are cached by Clark's nutcrackers (Tomback et al. 2001a), suggesting these mast events could have provided the seed source for the abundance of recently emerged whitebark pine seedlings we observed. However, the advanced regeneration we documented in the form of saplings indicates that there have been multiple episodes of successful whitebark pine establishment at our sites over the recent past.

Abundant seed production would do little for regeneration without the availability of sites suitable for regeneration. While whitebark pine regeneration is often associated with

Table 3. Whitebark pine (*Pinus albicaulis*) regeneration rates, blister rust rates on alive trees (% BR), frequency of mountain pine beetle-killed whitebark pine (PIAL-MPB), minimum stand age (Min Age), time since cohort establishment (TSC), and decadal disturbance frequency (DDF) for each mountain range.

	Regeneration rates of *Pinus albicaulis*						
Range	Seedlings/ha	Saplings/ha	% BR	PIAL-MPB	Min Age	TSC	DDF
Cascades	1233±326	703±93	0.11±0.03	0.67±0.06	324±46	137±24	0.12±0.05
Gravelly Range	375±158	467±113	0.63±0.07	0.78±0.06	290±50	124±23	0.09±0.02
Paulina Peak	6086±2113	840±160	0.06±0.03	0.70±0.10	311±28	163±44	0.11±0.03
Pioneer Mountains	3107±1072	866±189	0.41±0.06	0.59±0.10	453±44	279±62	0.08±0.01
Salmon River Mountains	92±29	252±43	0.61±0.10	0.67±0.09	242±31	143±34	0.13±0.01
Wallowa Mountains	215±108	149±33	0.37±0.10	0.34±0.07	442±59	225±49	0.04±0.01

Note: Values are means ± 1 SD.

Table 4. Results of Kruskal–Wallis' one-way analysis of variance by ranks test for differences in whitebark pine seedling and sapling densities among mountain ranges.

		Seedlings			Saplings		
Range	*n*	Median	Mean rank	*Z*	Median	Mean rank	*Z*
Cascades	6	1000	42.6	1.79	720	43.9	1.98
Gravelly Range	8	200	25.1	–0.95	470	31.5	0.17
Paulina Peak	7	6700	49.0	2.98	920	45.2	2.37
Pioneer Mountains	14	900	42.7	2.99	600	42.3	2.88
Salmon River Mountains	12	100	16.1	–3.20	220	20.0	–2.34
Wallowa Mountains	13	100	18.5	–2.81	100	12.8	–4.13

recently burned or otherwise disturbed settings, our results are more similar to a recent study in British Columbia that found comparable densities of whitebark pine seedlings between recently burned sites and nearby undisturbed settings (Moody 2006). This suggests that whitebark pine is not limited to the role of a postfire pioneer species at our study sites even in the presence of more shade-tolerant species, such as subalpine fir and mountain hemlock.

The mountain ranges with the lowest densities of regeneration illustrate contrasting mechanisms behind these patterns. The stands in the Wallowa Mountains that we examined were quite open in terms of stand density and total basal area and would seem to provide ample locations for whitebark pine regeneration (Tomback 1982), yet regeneration levels were relatively low compared with other sites. The Wallowa Mountains exhibited the lowest mean rates of blister rust infection, mountain pine beetle activity, and disturbance frequencies of the ranges included in our study and had a corresponding greater importance of subalpine fir, possibly suggesting that these stands were successionally advanced in the absence of disturbance. The inverse relationship between subalpine fir and whitebark pine regeneration identified in our correlation analyses and CAP suggests that subalpine fir may limit whitebark pine regeneration through resource competition and shading on suitable sites (Keane et al. 1990). Therefore, the lower rates of regeneration in the Wallowa Mountains may be the result of advancing succession due to a general lack of disturbance.

The Salmon River Mountains are climatically similar to the Wallowa Mountains and also exhibit low levels of regeneration. In the case of the Salmon River Mountains, however, the low levels of regeneration and greater importance of subalpine fir are more likely the result of the compounded effects of multiple recent disturbances. Most stands in the Salmon River Mountains experienced high levels of mountain pine beetle-related mortality from the late 1980s to the present in addition to two large fires that burned across two sites in 1989 and three sites in 1994. Large, infrequent disturbances that occur in close temporal succession such as these events often have different ecological effects than either disturbance alone (Paine et al. 1998). As a result, whereas whitebark pine regeneration was positively correlated with mortality caused by mountain pine beetle elsewhere in our study area, the fires that burned following mountain pine beetle activity at these sites may have killed whatever whitebark pine regeneration was alive at the time while also killing nearby seed sources that may have survived the mountain pine beetle outbreaks. In contrast, whitebark pine regeneration was greater in the mountain ranges with moderate disturbance frequencies.

These results suggest a disturbance-frequency-dependent pattern in whitebark pine regeneration that may have important implications for the ecological response of whitebark pine communities across the species range. In particular, the low levels of whitebark pine regeneration at sites affected by multiple closely timed disturbance events is a foreboding pattern in the context of expanding mountain pine beetle outbreaks (Raffa et al. 2008) and increasing mid- and high-elevation fire activity in the western United States (Westerling et al. 2006).

The relationships between whitebark pine regeneration and biophysical site characteristics

Similar to other tree species that exist across the gradient from lower subalpine forests to upper subalpine forests, patterns in whitebark pine regeneration reflect both local-scale factors as well as the overall dominant environmental gradients of regional climate and changing elevation (e.g.,

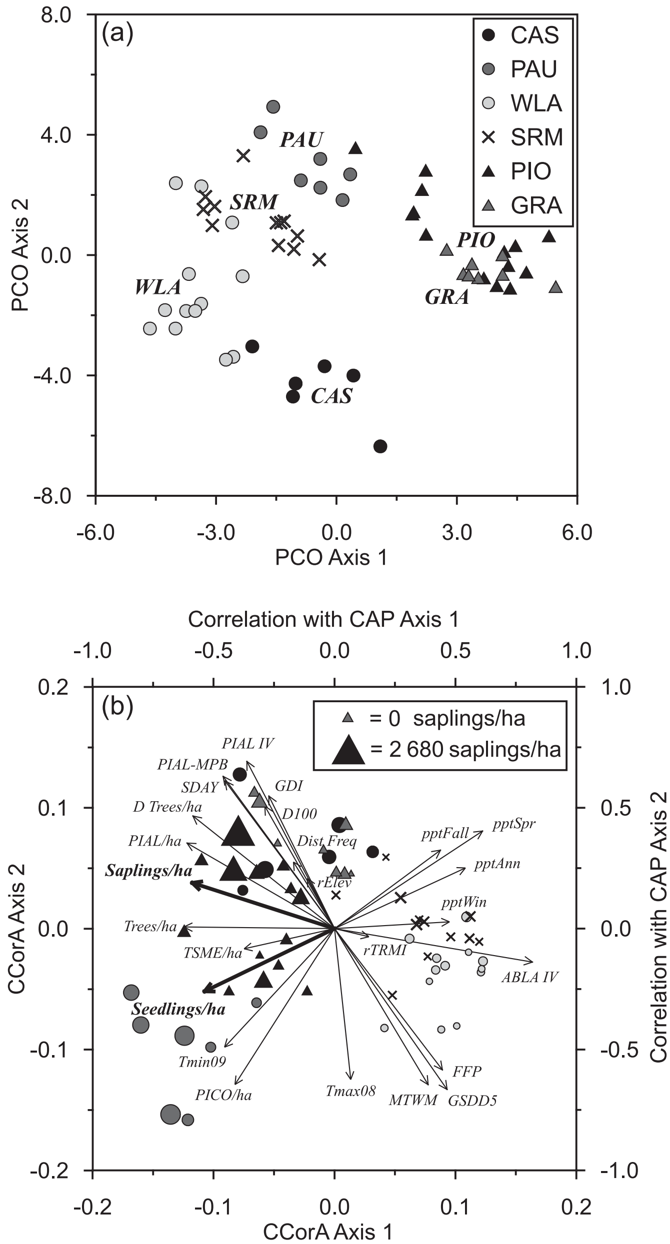
Table 5. Significant Kendall's τ rank correlations between biophysical site characteristics and whitebark pine seedling and sapling densities, and correlations between biophysical site characteristics and axes 1 and 2 of the Canonical Correlation Analysis (CCorA).

	Kendall's τ		CCorA	
Variable	τ	P	Axis 1	Axis 2
Seedlings				
ABLA IV	–0.43	<0.001		
Canopy trees/ha	0.42	<0.001		
Alive canopy PIAL/ha	0.39	<0.001		
pptFall	–0.28	0.001		
MTWM	–0.28	0.001		
PICO/ha	0.28	0.001		
Dead trees/ha	0.27	0.001		
pptSpr	–0.27	0.001		
GSDD5	–0.24	0.004		
SDAY	0.22	0.006		
rElev	0.22	0.007		
PIAL-MPB	0.19	0.015		
GDI	0.17	0.030		
Saplings				
ABLA IV	–0.46	<0.001	0.819	–0.139
PIAL/ha	0.42	<0.001	–0.610	0.354
Dead trees/ha	0.40	<0.001	–0.585	0.466
GSDD5	–0.37	<0.001	0.483	–0.665
MTWM	–0.35	<0.001	0.387	–0.645
PIAL-MPB	0.34	<0.001	–0.460	0.630
SDAY	0.33	<0.001	–0.452	0.610
Trees/ha	0.32	<0.001	–0.622	0.007
GDI	0.30	<0.001	–0.271	0.548
FFP	–0.30	<0.001	0.447	–0.584
PIAL IV	0.28	0.001	–0.362	0.692
pptSpr	–0.26	0.002	0.613	0.404
pptFall	–0.25	0.003	0.439	0.325
D100	0.25	0.003	–0.288	0.507
pptAnn	–0.24	0.003	0.541	0.251
rElev	0.23	0.004	–0.112	0.207
PICO/ha	0.22	0.008	–0.411	–0.645
pptWin	–0.21	0.008	0.475	0.028
Dist Freq	0.19	0.017	–0.169	0.276
Tmax08	–0.16	0.035	0.067	–0.625
TSME/ha	0.16	0.037	–0.374	–0.083
Tmin09	0.16	0.038	–0.454	–0.490
rTRMI	–0.15	0.047	0.142	–0.034

Note: In many cases significant correlations were identified between closely related variables, such as monthly and seasonal precipitation variables or variations on forest metrics (e.g., relative frequency, relative basal area, and importance values). In these cases, we listed the single variable with the strongest relationship. Species codes are as follows: ABLA, *Abies lasiocarpa*; PIAL, *Pinus albicaulis*; PICO, *Pinus contorta*; TSME, *Tsuga mertensiana*. For an explanation of most of the variable codes see Table 2; others are as follows: PIAL-MPB, *Pinus albicaulis* killed by mountain pine beetle; pptFall, mean fall precipitation; pptSpr, mean spring precipitation; pptAnn, mean annual precipitation; pptWin, mean winter precipitation; Dist Freq, disturbance frequency; Tmax, mean monthly maximum temperature; Tmin, mean monthly minimum temperature.

Pollmann and Veblen 2004). The relationships between whitebark pine regeneration and biophysical site characteristics that we identified agreed well with the known distribution of whitebark pine and the niche it occupies (Rehfeldt et al. 2008). The strong correspondence between axis 1 of our CCorA and several measures of seasonal and annual temperature and moisture availability (Fig. 2) illustrates whitebark pine's ability to regenerate in cold, dry environments and provides a mechanism for whitebark pine dominance of timberline communities and xeric sites in the subalpine forests

Fig. 2. Ordinations of the biophysical variables significantly correlated to whitebark pine sapling abundance using (*a*) Principal Coordinates analysis (PCO) and (*b*) Canonical Correlation Analysis (CCorA). Site codes in panel *a* indicate the approximate centroid of each range in ordination space. Symbols in panel *b* are scaled by sapling density. The correlations of the original variables with the two canonical axes are represented by vectors in panel *b*. Legend abbreviations in panel *a*: CAS, Cascade Range; PAU, Paulina Peak; WLA, Wallowa Mountains; SRM, Salmon River Mountains; PIO, Pioneer Mountains; GRA, Gravelly Range. For an explanation of the abbreviations in panel *b*, see Table 2 and the Note in Table 5.



of the Northern Rockies and Pacific Northwest (Arno 2001). The differing relationships between seedling and sapling densities and CCorA axis 2 potentially illustrate the differential effects of the biophysical environment on whitebark pine germination success versus successful establishment.

McCaughey (1990) found that while warm temperatures were important for facilitating whitebark pine seed germination, some of the most common causes of death among emerging whitebark pine seedlings were heat scorch and drought. Similarly, Moody (2006) found that whitebark pine seedlings were more abundant following longer growing seasons contingent on there being sufficient moisture available throughout the season. These relationships seem to play out in our study area, with the greater seedling densities found at the sites with longer, warmer growing seasons compared with the greater sapling densities at the colder sites within the context of the overall gradient of axis 1 toward colder sites in general. The implications of this are that while seedlings are more likely to emerge at warmer sites, they are also more likely to suffer heat damage and higher mortality rates, and therefore seedlings on cooler sites with lower initial seedling success have higher sapling recruitment rates (Tomback et al. 1993; McCaughey et al. 2009). This differential response likely affected the spatial patterning and structure of whitebark pine communities in the past, and may play an important role in the future dynamics of whitebark pine forests in the context of global warming.

The strong inverse relationship between whitebark pine seedling and sapling density and the importance of subalpine fir at our sites provides an interesting illustration of the complexity of regeneration dynamics in whitebark pine communities. Subalpine fir is more tolerant of shade than whitebark pine but less tolerant of drought and winter desiccation, and often depends on whitebark pine to become established at harsh sites (Callaway 1998). In moderate sites, however, subalpine fir often dominates the understory of whitebark pine forests as they age and transition into later successional stages (Kipfmueller and Kupfer 2005). Indeed, subalpine fir was the dominant understory species (based on importance values) in 22 of the 35 sites in our study at which it occurred, yet neither subalpine fir importance nor whitebark pine regeneration densities showed a significant relationship to stand age (Larson 2009), and the presence of recently emerged and advanced regeneration at these sites suggests that whitebark pine regeneration was not yet precluded from these stands. Campbell and Antos (2003) observed similar patterns of whitebark pine regeneration throughout the history of several stands in a chronosequence study of succession in whitebark pine – subalpine fir forests in British Columbia. Whitebark pine regeneration densities were significantly different between the landscapes included in our study, and these patterns may not hold true across the entire range of whitebark pine, but the presence of whitebark pine regeneration in later-successional subalpine forests may play an important role in the persistence of whitebark pine on the modern landscape, albeit at lower densities.

The role of mountain pine beetles in whitebark pine regeneration

The declines in whitebark pine forests due to blister rust are likely unprecedented, as this disease is a novel disturbance with respect to the recent evolutionary history of whitebark pine, and fire suppression activities are potentially causing shifts of many whitebark pine forests outside of their natural range of variability in terms of structure and composition (Murray et al. 2000; Larson et al. 2009). In contrast, mountain pine beetle has been a disturbance agent in whitebark pine forests for millennia (Brunelle et al. 2008), and while recent warming may have exacerbated current mountain pine beetle outbreaks (Hicke et al. 2006), this warmth may not be unprecedented in the evolutionary history of whitebark pine in North America when compared with the increased seasonality and temperature extremes associated with periods such as the Holocene Solar Optimum (Ruddiman 2000). It would therefore make sense if the tree species were adapted to occasional severe and widespread mountain pine beetle outbreaks similar to recent beetle epidemics. The strong relationship between rates of whitebark pine killed by mountain pine beetle and whitebark pine regeneration density indicate that stand-scale gap-phase dynamics may be one response of whitebark pine forests to mountain pine beetle outbreaks.

Gap dynamics have been extensively studied in forest ecosystems worldwide (e.g., Platt and Strong 1989 and papers mentioned therein), yet while the patchy mortality of most mountain pine beetle outbreaks creates numerous forest openings and canopy gaps of varying sizes (Raffa et al. 2008), relatively little research has examined how these disturbances may influence whitebark pine regeneration. In part, this is due to a broad focus on the historical role of fire as the dominant disturbance process in most forest types of western North America (Keane et al. 2002), including whitebark pine communities (Keane 2001). It also reflects the uncertainty and relatively limited data available on the long-term role of mountain pine beetles in whitebark pine ecosystems. Researchers have observed whitebark pine regeneration under canopies killed by mountain pine beetle in the past (Ciesla and Furniss 1975), and while research suggests that warming temperatures are enabling mountain pine beetle outbreaks to reach whitebark pine forests in climatic settings that were previously too harsh to support large beetle populations (Hicke et al. 2006), Brunelle et al. (2008) found evidence that mountain pine beetle outbreaks have occurred in whitebark pine ecosystems since at least the mid Holocene. In other forest types, mountain pine beetle outbreaks act as secondary disturbances with strong influences on patterns in stand development (Sibold et al. 2007). Our research adds to this growing literature by indicating that mortality caused by mountain pine beetle in whitebark pine forests can serve as an effective mechanism for creating canopy gaps and forest openings that appear to be attractive seed caching areas for Clark's nutcrackers (Hutchins and Lanner 1982; Tomback 1982) and are suitable sites for whitebark pine regeneration. Additionally, the thinner canopy and physical shelter provided to seedlings and saplings by the presence of standing snags following mountain pine beetle outbreaks is conducive to higher survival rates for whitebark pine regeneration in these sites (McCaughey et al. 2009).

Management implications

The existence of landscape-scale relationships between whitebark pine regeneration and the biophysical environment offer several opportunities to increase the efficacy of management and restoration activities in whitebark pine communities. Outplanting of blister rust-resistant seedlings and saplings is an expensive and labor-intensive endeavor (Hoff et al. 2001) but is considered one of the more effective strategies for managing whitebark pine in the presence of white pine blister rust (Schoettle and Sniezko 2007). In the context of our research and other recent studies (McCaughey et al. 2009), targeting stands in colder, drier settings for outplanting may give the planted whitebark pine the greatest chance of establishment and eventual maturation. Planting beneath canopies of mature whitebark pine experiencing an active mountain pine beetle outbreak may also be an effective approach, as the resources made available by mountain pine beetle disturbances appear to provide an optimal environment for whitebark pine regeneration. Additionally, stands where the mature trees are killed by mountain pine beetle may be less susceptible to future outbreaks because of the dearth of larger, living trees that are the preferred host of beetle infestations (Amman 1972) and, depending on the susceptibility of the landscape to fire, offer the longest potential disturbance-free growing period for the planted trees.

The advanced natural regeneration we observed in gaps created by mountain pine beetle may also serve as an important asset to management aimed at increasing blister rust resistance in whitebark pine. Blister rust infection levels across our study area were moderate relative to other regions (Smith and Hoffman 2000; Smith et al. 2008), yet the susceptibility of early and advanced regeneration to white pine blister rust (Tomback et al. 1995) may result in differential mortality among the regeneration we observed with a greater proportion of surviving seedlings and saplings representing rust-resistant genotypes. Rust resistance has been documented to increase rapidly over only a few generations (Hoff et al. 2001), and the greater levels of regeneration found at our study sites affected by 20th century mountain pine beetle outbreaks may act as a catalyst for the development of white pine blister rust resistance in whitebark pine. Natural regeneration in this context may provide a key mechanism for this foundation species to adapt to its changing environment and should be closely monitored, as it could be a critical step in maintaining the presence of whitebark pine communities in western North America.

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